of overlaps. It expresses the fact that the logical relation between states is preserved with time.

4.5 THE HAMILTONIAN

In the study of classical mechanics, we became familiar with the idea of an incremental change in time. Quantum mechanics is no different in this respect: we may build up finite time intervals by combining many infinitesimal intervals. Doing so will lead to a differential equation for the evolution of the state-vector. To that end, we replace the time interval t with an infinitesimal time interval ε and consider the time-evolution operator for this small interval.

There are two principles that go into the study of incremental changes. The first principle is unitarity:

$$\mathbf{U}^{\dagger}(\epsilon)\mathbf{U}(\epsilon) = I. \tag{4.5}$$

The second principle is continuity. This means that the state-vector changes smoothly. To make this precise, first consider the case in which ε is zero. It should be obvious that in this case the time-evolution operator is merely the unit operator I. Continuity means that when ε is very small, $\mathbf{U}(\varepsilon)$ is close to the unit operator, differing from it by something of order ε. Thus, we write

$$\mathbf{U}(\epsilon) = I - i\epsilon\mathbf{H}. \tag{4.6}$$

You may wonder why I put a minus sign and an i in front of $\mathbf{H}$. These factors are completely arbitrary at this stage. In other words, they are a convention that has no content. I used them with an eye toward the future, when we will recognize $\mathbf{H}$ as something familiar from classical physics.

We will also need an expression for $\mathbf{U}^{\dagger}$. Remembering that Hermitian conjugation requires the complex conjugation of coefficients, we find that